

EDUCATION

- **Purdue University – West Lafayette, IN, USA** **2023 – Present**
Ph.D. in Mechanical Engineering
- **University of Michigan – Ann Arbor, MI, USA** **2020 – 2022**
M.S. in Robotics
- **Purdue University – West Lafayette, IN, USA** **2014 – 2020**
B.S.E. in Mechanical Engineering

RESEARCH EXPERIENCE

- **School of Mechanical Engineering, Purdue University – West Lafayette, IN, USA** **8/2023 – Present**
Graduate Research Assistant
- **Department of Mechanical Engineering, University of Michigan – Ann Arbor, MI, USA** **6/2022 – 5/2023**
Research Associate
- **Robotics Department, University of Michigan – Ann Arbor, MI, USA** **1/2021 – 5/2022**
Graduate Research Assistant
- **School of Mechanical Engineering, Purdue University – West Lafayette, IN, USA** **5/2019 – 5/2020**
Undergraduate Research Assistant
- **Polytechnic Institute, Purdue University – West Lafayette, IN, USA** **5/2015 – 5/2016**
Undergraduate Research Assistant

TEACHING EXPERIENCE

- **Purdue University – West Lafayette, IN, USA** **1/2026 – Present**
Teaching Assistant: Instructed ML tutorials for graduate students (ME597) “Industrial IoT Implementation for Smart Manufacturing”
- **Purdue University – West Lafayette, IN, USA** **1/2015 – 5/2025**
Teaching Assistant: Instructed ML tutorials for graduate students (ME597) “Industrial IoT Implementation for Smart Manufacturing”

RESEARCH INTERESTS

- Robotics and Automation
- Computer vision and Artificial intelligence
- Cyber-physical system
- Smart manufacturing

MENTORING, OUTREACH, AND LEADERSHIP ACTIVITIES

- **Summer Undergraduate Research Fellowships (SURF) program**
Purdue University – West Lafayette, IN, USA
 - 01. Yoonseo Lee (B.S.E in Mechanical Engineering) **2025**
- Topic: Safety-aware high-dimensional motion planning
 - 02. Richard Li (B.S. in Computer Science) **2024**
- Topic: Optimization-based robotic grasp detection
- **Mentoring Undergraduate and Master Students**
School of Mechanical Engineering, Purdue University – West Lafayette, IN, USA

01. Trevor Terrill Kates (B.S.E in Mechanical Engineering) - Topic: Robot-integrated cell-based manufacturing toward physical AI	2026 – Present
02. Daniel Sungwoo Bae (M.S in Mechanical Engineering) - Topic: AI-integrated autonomous sealing robot	2026 – Present
03. Junho Sohn (M.S in Mechanical Engineering) - Topic: Interpretable AI for human motion analysis in manufacturing tasks <i>(Journal submission)</i> Sohn, J., Lee, H., Lee, D., Torres, A., Neira-García, J., Zhou, Q., Abdollahi, M., Rastgaar, M., & Jun, M. (2026). HANDS: Unsupervised Segmentation for Interpretable Identification of Human Motion Primitives in Manufacturing Task. <i>Journal of Manufacturing Systems</i> .	2025 – Present
04. Yoonseo Lee (B.S.E in Mechanical Engineering) - Topic: Safety-aware high-dimensional motion planning	2025
05. Evin Lugo Li (B.S.E in Mechanical Engineering) - Topic: Adaptive grasp planning for suction and two-finger grippers <i>(Conference proceeding)</i> Lee, H., Choi, Y. W., Lugo, E., Lee, J., Lee, S. W., & Jun, M. B. G. (2025, June). GMGP: Generalized Model for Grasp Planning of Vacuum and Parallel Jaw Grippers. In <i>International Manufacturing Science and Engineering Conference</i> (Vol. 89022, p. V002T16A008). American Society of Mechanical Engineers.	2024 – 2025
● Multidisciplinary Design Program (MDP) University of Michigan – Ann Arbor, MI, USA	
01. Yuhang Ning (M.S in Electrical and Computer Engineering) - Topic: Autonomous environment mapping for periodically reconfigured environments	2022
02. Jiwen Jia (M.S in Electrical and Computer Engineering) - Topic: Visual servo control design for a mobile manipulator	2021

FELLOWSHIPS, HONORS, AND AWARDS

● Best Paper Award (2 nd) International Manufacturing Science and Engineering Conference (MSEC) 2026, State College, PA, USA	2026
● Best Poster Award U.S. – Korea Conference (UKC) 2025, Atlanta, GA, USA	2025
● <i>Italian Machine Tool Technology Award</i> Italian Trade Agency, Italy	2025
● <i>NSF Student Travel Support Award</i> ASME International Manufacturing Science and Engineering Conference (MSEC) 2025 SME North American Manufacturing Research Conference (NAMRC) 53, Clemson University – Greenville, SC, USA	2025
● <i>Young Professional NET Award</i> The Korean Federation of Science and Technology <i>(Conference presentation)</i> Sim, Y.*., Lee, J., Lee, H., Han, C., Lee, H., & Jun, M. (2025, August 17-20). <i>I2DTM: Immersive and Interactive Digital Twin Metaverse Framework</i> [Conference Presentation]. International Design Engineering Technical Conference & Computers and Information in Engineering Conference 2025, Anaheim, CA, United States. <i>(Journal publication)</i> Jeon, J., Sim, Y., Lee, H., Han, C., Yun, D., Kim, E., Nagendra, S. L., Jun, M., Kim, Y., Lee, S., & Lee, J.* (2025). ChatCNC: Conversational Machine Monitoring via Large Language Model and Real-Time Data Retrieval Augmented Generation. <i>Journal of Manufacturing Systems</i> , 79, 504-514.	2024 – 2025
● <i>Best Poster Award</i>	2024

- *Winkelman Fellowship* 2023 – 2024
Purdue University – West Lafayette, IN, USA
- *DSCD and AACC Student Travel Awards* 2022
The 2022 Modeling Estimation and Control Conference (MECC), Jersey City, NJ, USA.
- *Multidisciplinary Design Program (MDP) Summer Fellowship* 2021
University of Michigan – Ann Arbor, MI, USA

JOURNAL PUBLICATIONS

- | ‡: Equal contribution | *: Corresponding author |
|-----------------------|---|
| 09. | Lee, H. , Han, C., Gabor, T., Sim, Y., Akin, S., Jun, M., & Lee, J.* (2026). Cold spray-based secure and unique product identification with neural encoding: A full-stack framework for scalable authentication in manufacturing. <i>Journal of Intelligent Manufacturing</i> , doi: https://doi.org/10.1007/s10845-026-02864-w |
| 08. | Sim, Y., Kim, E.* Lee, H. , Sohn, J., & Jun, M. (2025). Structured Sound Data Augmentation Based on Spectral Analysis for Machine Tool Operational State Monitoring. <i>The International Journal of Advanced Manufacturing Technology</i> , 141, 1705-1719. doi: https://doi.org/10.1007/s00170-025-16797-w |
| 07. | Kim, E., Lee, H. , Sim, Y., Lee, J., & Jun, M.* (2025). Overcoming Sparse Run-to-Failure Data Challenges in Manufacturing: A Contrastive Mixer Framework for Remaining Useful Life Prediction. <i>CIRP Annals – Manufacturing Technology</i> , 74, 579-583. doi: https://doi.org/10.1016/j.cirp.2025.04.082 |
| 06. | Han, C., Gabor, T., Lee, H. , Lee, J., Akin, S., & Jun, M.* (2025). Pulsed Cold Spray System for Physical Unclonable Function Generation. <i>Procedia CIRP</i> , 137, 176-180. doi: https://doi.org/10.1016/j.procir.2025.02.269 |
| 05. | Jeon, J., Sim, Y., Lee, H. , Han, C., Yun, D., Kim, E., Nagendra, S. L., Jun, M., Kim, Y., Lee, S., & Lee, J.* (2025). ChatCNC: Conversational Machine Monitoring via Large Language Model and Real-Time Data Retrieval Augmented Generation. <i>Journal of Manufacturing Systems</i> , 79, 504-514. doi: https://doi.org/10.1016/j.jmsy.2025.01.01 (#1 in Operations Research & Management Science ^{OOIR 2025}) |
| 04. | Lee, H.* , Toner, T., Tilbury, D., & Barton, K. (2025). GraspMixer: Hybrid of Contact Surface Sampling and Grasp Feature Mixing for Grasp Synthesis. <i>IEEE Transactions on Automation Science and Engineering</i> , 22, 10920- 10936. doi: 10.1109/TASE.2025.3530795 (#12 in Automation & Control System ^{OOIR 2025}) |
| 03. | Lee, J., Akin, S.* Lee, H. , Kim, E., Nam, J., Song, K., & Jun, M.* (2024). A stethoscope-guided interpretable deep learning framework for powder flow diagnosis in cold spray additive manufacturing. <i>Manufacturing Letters</i> , 41, 1515-1525. doi: https://doi.org/10.1016/j.mfglet.2024.09.178 |
| 02. | Han, C., Lee, J., Lee, H. , Sim, Y., Jeon, J., & Jun, M.* (2024). Zero-shot Autonomous Robot Manipulation via Natural Language. <i>Manufacturing Letters</i> , 42, 16-20. doi: https://doi.org/10.1016/j.mfglet.2024.09.067 |
| 01. | Lee, H.* , Toner, T., Tilbury, D., & Barton, K. (2022). Multi-sensor aided deep pose tracking. <i>IFAC-PapersOnLine</i> , 55(37), 326-332. doi: https://doi.org/10.1016/j.ifacol.2022.11.205 |

JOURNAL IN PROCESS

- | ‡: Equal contribution | *: Corresponding author |
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| 09. | (In preparation) Lee, H. , Sim, Y., Lee, H., Lee, K., Jun, M., & Lee, J.* (2026). BRIDGES: Bi-directional Robotic Integration with Digital-twin for Generalized, Efficient, and Safe Automation in Manufacturing. |

08. (In preparation) Sim, Y., **Lee, H.**, Lee, H., Paek, S., Lee, J., Yoon, B., & Jun, M.* (2026). MMRL: A Multi-Modal Avatar based Remote Interaction and Localization Framework for Digital Twin Cross-Validation in Smart Manufacturing.
07. (Accepted in Jun. 2026) Sim, Y.*, Kim, E., Sohn, J., Barr, J., **Lee, H.**, Beach, B., Jun, M. & Shakouri, A. (2026). Generative Latent Space Modeling of Stamping Sound for Anomaly Detection and Predictive Maintenance. *Journal of Manufacturing Science and Engineering*. (2nd Best Paper Award)
06. (Under review, submitted in Mar. 2026) **Lee, H.**, Sim, Y., Lee, H., Yoon, B., Kim, C., & Jun, M.* (2025). PRISM: Progressive Refinement via Informed Sampling for Multi-belief Object Pose Validation. *Robotics and Computer-Integrated Manufacturing*.
05. (Under review, submitted in Mar. 2026) **Lee, H.**, Sim, Y., Yoon, B., Lee, H., Park, H., Jang, J., Kim, C., & Jun, M.* (2026). Toward provably safe AI-driven perception and motion planning for robotic manufacturing: A review. *Journal of Manufacturing Systems*.
04. (Under review, submitted in Jan. 2026) Sohn, J., **Lee, H.**, Lee, D., Torres, A., Neira-García, J., Zhou, Q., Abdollahi, M., Rastgaar, M., & Jun, M.* (2026). HANDS: Unsupervised Segmentation for Interpretable Identification of Human Motion Primitives in Manufacturing Task. *Journal of Intelligent Manufacturing*.
03. (Under review, submitted in Jan. 2026) Jun, M.*, Sim, Y., Kim, E., **Lee, H.** (2026). Open-world condition recognition for discrete manufacturing. *CIRP Journal of Manufacturing Science and Technology*.
02. (Under review, submitted in Dec. 2025) Nam, J., **Lee, H.**, Han, C., Athinarayanan, R., Jun, M., & J, Lee.* (2025). C³AM: Conversational CAD/CAM Automation via Multimodal Large Language Model Agents. *Journal of Intelligent Manufacturing*.
01. (In second round of review) **Lee, H.**, Sim, Y., Han, C., Jun, M., & Lee, J.* (2025). GRAVITON: Generalized Resampling Algorithm for Visual Inference in Target Object Localization. *Computers in Industry*.

PEER-REVIEWED CONFERENCE PROCEEDINGS

‡: Equal contribution

*: Corresponding author

12. (In preparation) Lee, H., Sim, Y., **Lee, H.**, Yoon, B., Lee, J., Garcia, J. E. N., Rastgaar, M., Jang, J., & Jun, M.* (2026, November). High-Precision 3D Trajectory Estimation Using Wire Encoder based Module with String Dynamic Model Compensation. In 2026 International Mechanical Engineering Congress and Exposition (IMECE). American Society of Mechanical Engineers.
11. (Under review, submitted in Apr. 2026) Jun, M.*, Sim, Y., **Lee, H.**, Kim, E., & Shakouri, A. (2026, October 11-16). *Acoustic Fingerprinting for Online Condition Monitoring in Progressive Stamping*. International Conference on the Technology of Plasticity 2026, Jeju Island, Republic of Korea.
10. (Accepted in Jun. 2026) Yoon, B., **Lee, H.**, Sim, Y., & Jun, M.* (2026, September). LiDAR-Integrated Coarse-to-Fine Optimization for Geometrically Consistent Triangle Splatting from Mobile Robots. In 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE.
09. (Accepted in May. 2026) Lee, H., Sim, Y., Yoon, B., **Lee, H.**, & Jun, M.* (2026, June 2-3). *Development of Scalable FSR Sensor Array-based Human Footprint Acquisition Module for Manufacturing Automation*. Excellence in Manufacturing and Operations (XMO) Industrial Seminar 2026, West Lafayette, IN, United States.
08. (Accepted in May. 2026) Sim, Y., Lee, Y., Yother, E., Lee, D., **Lee, H.**, Yoon, B., & Jun, M.* (2026, June 2-3). *Real-time Two-stage Machine Sound Monitoring for Wire-EDM State Recognition*. Excellence in Manufacturing and Operations (XMO) Industrial Seminar 2026, West Lafayette, IN, United States.

07. (Accepted in May. 2026) Bae, D., Han, C., Lee, H., **Lee, H.**, & Jun, M.* (2026, June 2-3). *RGB-D Object Pose Estimation via SAM2 Segmentation and Coarse-to-Fine Point Cloud Registration*. Excellence in Manufacturing and Operations (XMO) Industrial Seminar 2026, West Lafayette, IN, United States.
06. (Accepted in Jan. 2026) Nam, J.*, **Lee, H.**, Han, C., Athinarayanan, R., Jun, M., & J, Lee. (2026, June). *Gen-Cad/cam: Generative and Interactive Cad/cam Framework for Simulation-Informed Adaptive Maintenance*. In *International Manufacturing Science and Engineering Conference*. American Society of Mechanical Engineers.
05. (Under review, submitted in Jan. 2026) Han, C., Sim, Y., Jung, H., Lee, J., **Lee, H.**, Kang, Y., Woo, S., Kim, G., Park, H., & Jun, M.* (2026, July). *IMPACT: Industrial Machine Perception via Acoustic Cognitive Transformer*. In *Knowledge Discovery and Data Mining 2026 (KDD)*. ACM.
04. **Lee, H.***, Sim, Y., Han, C., Lee, J., Bera, A., Kim, C., & Jun, M. B. G. (2025, October). *EASEIR: Efficient and Adaptive Safe-set Estimation via Implicit Representation for High-dimensional Motion Planning*. In *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 11661-11668). IEEE. doi: [10.1109/IROS60139.2025.11246375](https://doi.org/10.1109/IROS60139.2025.11246375)
03. Sim, Y.*, Kim, E., Lee, J., **Lee, H.**, Yun, D., Myeong, N., Hwang, Y., Park, H., & Jun, M. B. G. (2025, June). *Sound Data Augmentation Using Frequency Response Superposition for Machine Tool Component Running State Recognition*. In *International Manufacturing Science and Engineering Conference* (Vol. 89022, p. V002T16A006). American Society of Mechanical Engineers. doi: <https://doi.org/10.1115/MSEC2025-155613>
02. **Lee, H.***, Choi, Y. W., Lugo, E., Lee, J., Lee, S. W., & Jun, M. B. G. (2025, June). *GMGP: Generalized Model for Grasp Planning of Vacuum and Parallel Jaw Grippers*. In *International Manufacturing Science and Engineering Conference* (Vol. 89022, p. V002T16A008). American Society of Mechanical Engineers. doi: <https://doi.org/10.1115/MSEC2025-155673> (NSF Student Travel Support Award)
01. Han, C.*, **Lee, H.**, Lee, J., Jun, M. B. G., & Yun, H. (2024, June). *A Study on Digital Twin for Autonomous Robotic Chip Removal Deep Learning Framework on CNC Machine*. In *International Manufacturing Science and Engineering Conference* (Vol. 88100, p. V001T02A003). American Society of Mechanical Engineers. doi: <https://doi.org/10.1115/MSEC2024-124383>

CONFERENCE PRESENTATIONS

‡: Presenter

*: Corresponding author

14. Jun, M.‡*, **Lee, H.**, Sim, Y., Lee, H., Yoon, B., & Bae, D. (2026, June 28- July 3). *Leveraging a Vision Foundation Model for Adaptable and Robust Object 3D Localization and Manipulation under Real Shop-Floor Environments*. International Conference on Precision Engineering and Sustainable Manufacturing 2026, Singapore.
13. Nam, J.*, **Lee, H.**, Han, C., Athinarayanan, R., Jun, M., & J, Lee. (2026, June 14-18). *Gen-Cad/cam: Generative and Interactive Cad/cam Framework for Simulation-Informed Adaptive Maintenance*. 2026 International Manufacturing Science and Engineering Conference, State College, PA, United States.
12. Sim, Y.*, Kim, E., Sohn, J., Barr, J., **Lee, H.**, Beach, B., Jun, M. & Shakouri, A. (2026, June 14-18). *Generative Latent Space Modeling of Stamping Sound for Anomaly Detection and Predictive Maintenance*. 2026 International Manufacturing Science and Engineering Conference, State College, PA, United States.
11. **Lee, H.‡***, Sim, Y., Lee, H., Yoon, B., Kim, C., & Jun, M. (2026, January 29-30). *PRISM: Progressive Refinement via Informed Sampling for Multi-belief Object Pose Estimation*. 2026 Institute for Control, Optimization and Network Student Research Conference, West Lafayette, IN, United States.

10. **Lee, H.^{†*}**, Sim, Y., Han, C., Lee, J., Bera, A., Kim, C., & Jun, M. B. G. (2025, October 19-25). *EASEIR: Efficient and Adaptive Safe-set Estimation via Implicit Representation for High-dimensional Motion Planning*. 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems, Hangzhou, China.
09. Kim, E., **Lee, H.**, Sim, Y., Lee, J., & Jun, M.^{†*} (2025, August 17-23). *Overcoming Sparse Run-to-Failure Data Challenges in Manufacturing: A Contrastive Mixer Framework for Remaining Useful Life Prediction*. 74th CIRP Annals – Manufacturing Technology, Stockholm, Sweden.
08. Sim, Y.^{†*}, Lee, J., **Lee, H.**, Han, C., Lee, H., & Jun, M. (2025, August 17-20). *I2DTM: Immersive and Interactive Digital Twin Metaverse Framework*. International Design Engineering Technical Conference & Computers and Information in Engineering Conference 2025, Anaheim, CA, United States.
07. **Lee, H.**, Sim, Y., Han, C., Lee, J., & Jun, M.^{†*} (2025, July 6-11). *ARMPASS: Autonomous Robot Motion Planning with Adaptive Safe Set*. International Conference on Precision Engineering and Sustainable Manufacturing 2025, Chiang Mai, Thailand.
06. Sim, Y.^{†*}, Kim, E., Lee, J., **Lee, H.**, Yun, D., Myeong, N., Hwang, Y., Park, H., & Jun, M. B. G. (2025, June 23-27). *Sound Data Augmentation Using Frequency Response Superposition for Machine Tool Component Running State Recognition*. International Manufacturing Science and Engineering Conference 2025, Greenville, SC, USA.
05. **Lee, H.^{†*}**, Choi, Y. W., Lugo, E., Lee, J., Lee, S. W., & Jun, M. B. G. (2025, June 23-27). *GMGP: Generalized Model for Grasp Planning of Vacuum and Parallel Jaw Grippers*. International Manufacturing Science and Engineering Conference 2025, Greenville, SC, USA.
04. Han, C.[†], Gabor, T., **Lee, H.**, Lee, J., Akin, S., & Jun, M.* (2025, June 1-4). *Pulsed Cold Spray System for Physical Unclonable Function Generation*. CIRP International Symposium on Electro Physical and Chemical Machining 2025, Vancouver, BC, Canada.
03. Jun, M.^{†*}, Lee, J., Kim, E., Sim, Y., Han, C., & **Lee, H.** (2024, July 7-12). *Generalized and Generative AI for Smart Manufacturing*. International Conference on Precision Engineering and Sustainable Manufacturing 2024, Busan, Republic of Korea.
02. Han, C.^{†*}, **Lee, H.**, Lee, J., Jun, M. B. G., & Yun, H. (2024, June 17-21). *A Study on Digital Twin for Autonomous Robotic Chip Removal Deep Learning Framework on CNC Machine*. International Manufacturing Science and Engineering Conference 2024. Knoxville, TN, USA.
01. **Lee, H.^{†*}**, Toner, T., Tilbury, D., & Barton, K. (2022, October 2-5). *Multi-sensor aided deep pose tracking*. Modeling, Estimation and Control Conference 2022, Jersey City, NJ, United States.

POSTER SESSIONS

09. Lee, H., Sim, Y., Yoon, B., **Lee, H.**, & Jun, M.* (2026, June 2-3). *Development of Scalable FSR Sensor Array-based Human Footprint Acquisition Module for Manufacturing Automation*. Excellence in Manufacturing and Operations (XMO) Industrial Seminar 2026, West Lafayette, IN, United States.
08. Sim, Y., Lee, Y., Yother, E., Lee, D., **Lee, H.**, Yoon, B., & Jun, M.* (2026, June 2-3). *Real-time Two-stage Machine Sound Monitoring for Wire-EDM State Recognition*. Excellence in Manufacturing and Operations (XMO) Industrial Seminar 2026, West Lafayette, IN, United States.
07. Bae, D., Han, C., Lee, H., **Lee, H.**, & Jun, M.* (2026, June 2-3). *RGB-D Object Pose Estimation via SAM2 Segmentation and Coarse-to-Fine Point Cloud Registration*. Excellence in Manufacturing and Operations (XMO) Industrial Seminar 2026, West Lafayette, IN, United States.
06. Han, C., Sim, Y., Lee, J., **Lee, H.**, & Jun, M. (2025, August 5-9). Industrial Audio Foundation for Agile Machine Listening. U.S. – Korea Conference 2025, Atlanta, GA, United States. (Best Poster Award)

05. Sim, Y., Lee, J., **Lee, H.**, Yun, D., Myeong, N., Hwang, Y., Park, H., Jun, M., & Kim, E. (2024, December 2). *Sound Data Augmentation Using Frequency Response Superposition for Machine Tool State Recognition*. Manufacturing And Materials Research Laboratories Symposium 2024, West Lafayette, IN, United States.
04. Lee, J., Jeon, J., Sim, Y., **Lee, H.**, Han, C., Yun, D., & Jun, M. (2024, October 15-16). *Generative AI Based Real-Time Human-Data Interaction for Smart Manufacturing*. Digital Twin for Manufacturing Sustainability, Safety, and Resilience 2024, West Lafayette, IN, United States. (**Best Poster Award**)
03. **Lee, H.**, Sim, Y., Lee, J., Han, C., & Jun, M. (2024, October 15-16). *Digital Twin & Vision Guided Autonomous Robotic Path Planning*. Digital Twin for Manufacturing Sustainability, Safety, and Resilience 2024, West Lafayette, IN, United States.
02. Lee, J., Kim, E., Han, C., **Lee, H.**, & Sim, Y. (2024, March 24-26). *Generalized and Generative Smart Manufacturing*. North Academy of Engineering Regional Meeting 2024, West Lafayette, IN, United States.
01. Han, C., **Lee, H.**, & Jun, M. (2024, May 24-26). *Robotics in Smart Manufacturing*. Congressional Artificial Intelligence Caucus 2024, Washington, D.C., United States.

PATENTS

01. Jun, M., Lee, J., **Lee, H.**, Akin, S., Han, C., Gabor, T., & Sim, Y. (2025, September 18). *Cold spray-enabled physically unclonable identifier and its spectral authentication via implicit neural representation* (U.S. Patent Application No. 64/033,490). U.S. Patent and Trademark Office.

PROFESSIONAL SOCIETY MEMBERSHIPS & SERVICES

- Reviewer of the IEEE Transactions on Automation Science and Engineering 2025 – Present
- Reviewer of the IEEE/RSJ International Conference on Intelligent Robots and Systems 2025 – Present
- Reviewer of the ASME Manufacturing Science and Engineering Conference 2025 – Present
- Member of the Institute of Electrical and Electronics Engineers (IEEE) 2025 – Present
- Member of the American Society of Mechanical Engineers (ASME) 2025 – Present
- Member of the Society of Manufacturing Engineers (SME) 2025 – Present
- Graduate Association for Manufacturing Excellence / Treasurer 2025 – Present
- Korean-American Scientists and Engineers Association (KSEA) / Member 2024 – Present

INDUSTRIAL PROJECTS

- **Genesem, Inc.** 2026 – Present
 - Development of a CAD-based part localization in a semiconductor equipment for part replacement.
- **Hitachi America, Ltd.** 2024 – Present
 - Developed a multi-modal feature extraction method to fuse data from a glove sensor and a camera.
 - Extended the feature extraction module to perform motion analysis and task segmentation of human operators during manufacturing processes.
- **BCI Solution, Inc** 2024 – 2025
 - Implemented a digital twin of FANUC CRX-10iA/L using MuJoCo physics simulator and Python.
 - Established bi-directional communication among FANUC CRX-10iA, its digital twin, and CESMII platform for automation and verification of core-making operations.

- **Korea Institute of Machinery and Materials (KIMM)** 2024 – Present
 - Developed a generalized sampling algorithm to extend a cutting edge object segmentation AI to 6-DOF object pose estimation method for machine tending.
 - Developed a hybrid computer vision pipeline to localize metal debris inside CNC machine that must be removed.

- **BCI Solution, Inc** 2023 – Present
 - Built a server-client system using Python and KAREL to communicate FANUC controller via external PC.
 - Integrated 6-DOF object pose estimation algorithm with FANUC controller via the server-client system for tool-path adaptation

- **Terra Drive Systems (TDS)** 2023 – 2023
 - Installed a stethoscope-based sound sensor and a mini PC to establish a data pipeline.
 - Developed a sound recognition AI model to monitor status of Makino a81 CNC machine.
 - Generated a web-based dashboard for intuitive data analysis and visualization.